Coordinate Geometry

Worksheet 2 - (Non-Calculator)

1. W15/qp12/q22

P is the point (1, –3) and Q is the point (7, 2).

a. Find the coordinates of the midpoint of PQ. [1]
b. Find the gradient of the line PQ. [1]
c. The line, L, with equation $2x - 5y = k$, passes through the point Q.
   i. Find the value of k. [1]
   ii. The line $x + Ay = 3$ is parallel to L. Find the value of A. [1]

2. S16/qp21/q25 (0580)

A is the point (4, 1) and B is the point (10, 15). Find the equation of the perpendicular bisector of the line AB. [6]

3. S19/qp11/q17

A line segment, AB, joins A (3, 2) to B (-1, 10).

a. Find the coordinates of the midpoint of AB. [1]
b. Find the equation of the perpendicular bisector of AB. [4]

4. S20/qp12/q24

P is the point (h, 7). P lies on the line $3y + 2x = 5$.

a. Find the value of h. [2]
b. Line L is perpendicular to the line $3y + 2x = 5$ and passes through P. Find the equation of line L. [4]

5. W20/qp11/q21

Line L is perpendicular to the line $y = -\frac{1}{2}x + 3$. Line L passes through the point (8, 9). Find the equation of line L. [3]

6. W20/qp22/q24 (0580)

A line from the point (2, 3) is perpendicular to the line $y = \frac{1}{3}x + 1$. The two lines meet at the point P. Find the coordinates of P. [5]

7. W20/qp23/q12 (0580)

A straight line, *l* , has equation $y = 5x + 12$.

a. Write down the gradient of line l. [1]
b. Find the coordinates of the point where line l crosses the x-axis. [2]
c. A line perpendicular to line l has gradient k. Find the value of k. [1]

8. S21/qp22/q17 (0580)

Find the gradient of the line that is perpendicular to the $3y = 4x - 5$. [2]

9. W21/qp21/q11 (0580)

Line L has equation $y = 4 - 5x$. Find the equation of a line that is perpendicular to line L and passes through the point (0, 6). [3]

10. W21/qp22/q15 (0580)

a. A is the point (3, 16) and B is the point (8, 31). Find the equation of the line that passes through A and B. Give your answer in the form $y = mx + c$. [3]
b. The line CD has equation $y = 0.5x - 11$. Find the gradient of a line that is perpendicular to the line CD. [1]

Answer Key:

1. (a) $(4, -\frac{1}{2})$ (b)$\frac{5}{6}$ (c) (i) 4 (ii) –2.5, or any equiv.

2. y= $-\frac{3}{7}x + 11$

3.(a) (1,6) (b) $2y = x + 11$

4.(a)-8 (b)$y = \frac{3}{2}x + 19$

5. y= $2x - 7$

6. (2.4, 1.8)

7.(a)5 (b) $(-\frac{12}{5}, 0)$ (c)$-\frac{1}{5}$

8.-0.75

9. $y = \frac{1}{5}\ x + 6$

10.(a)$y = 3x + 7$ (b) -2